

ABHIJITH SIVAPRASADAN

Mechanical / Thermal / Test Engineer | CFD · Instrumentation · CAD · Energy Systems
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PROFILE

Mechanical and sustainable energy engineer with a strong focus on thermo-fluid systems, test-rig engineering, CFD/CHT analysis, measurement-chain commissioning, CAD-supported design and technical documentation. Recent Master's thesis at Siemens Energy involved high-temperature gas-turbine test-rig hardware, pressure-loss and thermal-flow analysis, ANSYS Fluent simulations, Python post-processing, NI-DAQ/LabVIEW workflows and validation planning. Interested in Westinghouse opportunities in mechanical engineering, thermal hydraulics testing, development engineering, mechanical design/calculation support, quality-oriented documentation and energy/nuclear technology. Registered as a student for Autumn 2026, making both summer/internship and permanent early-career roles relevant.

WESTINGHOUSE-RELEVANT FIT

- Thermo-fluid and test-rig background from Siemens Energy: CFD/CHT, pressure-drop analysis, high-temperature hardware, instrumentation and validation planning.
- Mechanical engineering foundation: B.Tech Mechanical Engineering plus M.Sc. Sustainable Energy Engineering, with coursework in thermodynamics, heat transfer, fluid mechanics, thermal systems and energy technology.
- Testing and instrumentation exposure: NI-DAQ, NI MAX, LabVIEW, thermocouples, pressure sensors, measurement-chain checks and structured troubleshooting.
- CAD and design exposure: SolidWorks from bachelor-level mechanical design projects and SAE competitions; Siemens NX exposure in later engineering context.
- Documentation discipline: technical reports, assumptions/limitations, validation needs, KPI reporting, test-readiness notes and structured engineering communication.
- No direct nuclear industry experience yet, but strong motivation to learn nuclear procedures, safety culture, thermal-hydraulic testing, mechanical integrity, quality and documentation practices.
- Open to both early-career permanent roles and summer/internship/student roles in Västerås, including mechanical, thermal hydraulics, testing, development and documentation-related opportunities.

PROFESSIONAL EXPERIENCE

Test Engineer / Master Thesis Student

Siemens Energy AB, Fluid Dynamic Lab, Finspång, Sweden | Apr 2025 - Nov 2025

Thesis: Numerical Investigation of Steady-State Thermo-Fluid Performance of a Reducer for a High-Temperature Dynamic Pressure Sensor Calibration Rig.

- Worked in a high-temperature gas-turbine R&D test environment, connecting simulation, measurement readiness, test-rig hardware and engineering validation.
- Developed steady-state compressible CFD and conjugate heat-transfer models in ANSYS Fluent for thermal-fluid performance assessment.
- Evaluated pressure-loss coefficient, peak Mach number, wall-temperature behaviour and thermal-delivery indicators across operating cases.
- Performed mesh-independence checks, near-wall validation and sensitivity studies to improve confidence in simulation results.
- Used Python with pandas, NumPy and matplotlib for KPI extraction, case comparison, visualisation and repeatable post-processing.
- Commissioned and checked NI-DAQ measurement chains with thermocouples and static/dynamic pressure sensors using NI MAX and LabVIEW workflows.
- Documented boundary conditions, assumptions, limitations, validation needs and next-step experimental verification in a formal thesis and technical reports.
- Performed structured root-cause analysis of a heater failure during testing and proposed practical mitigation actions.

Energy Efficiency Intern

Alleima AB, Sandviken, Sweden / Remote | Dec 2024 - Jun 2025

- Developed an ISO 50001/EU EED-aligned energy performance mapping methodology for industrial sites.
- Worked with industrial utilities and process-energy systems including process heat, compressed air, heat recovery and metering readiness.
- Defined KPI/EnPI frameworks, load-driver normalisation logic, measurement plans and energy-performance reporting structures.
- Built Python analytical workflows to compare operating states, identify anomalies and support practical energy-efficiency improvement roadmaps.

Backend Engineer

QBBurst, Thiruvananthapuram, India | Sep 2021 - Jun 2023

- Developed backend APIs and services using Python/TypeScript-based workflows, Postman, Git and Docker.
- Worked with automated testing, structured debugging, issue reproduction, documentation and team-based software delivery.
- Built transferable skills in traceability, careful change control, reproducible workflows and technical communication.

SELECTED PROJECTS

High-Temperature Thermo-Fluid Test-Rig Thesis

Siemens Energy / KTH | ANSYS Fluent, Python, NI-DAQ, LabVIEW | 2025

- Modelled coupled flow and heat-transfer behaviour in gas-turbine test-rig hardware using CFD/CHT.
- Assessed pressure-loss behaviour, thermal performance, operating assumptions and design implications.
- Linked simulation outputs with measurement-chain readiness and validation planning for future experimental correlation.

Mechanical Design and SAE Mobility Projects

Bachelor-level projects | SolidWorks, component design, documentation, team delivery | 2018 - 2021

- Developed early CAD and mechanical design experience in SolidWorks through SAE Bicycle Design Competition and other bachelor-level mechanical design projects.
- Contributed to SAE BAJA and Tractor Design Competition work, including drivetrain concepts, component selection, transmission/specification research and technical documentation.
- Built practical understanding of assembly thinking, manufacturability, mechanical constraints, design iteration and team-based engineering delivery.

Distribution Grid and Renewable Integration Studies

KTH Course Project | Python-based power system workflow | 2025

- Analysed a CIGRE MV distribution grid using load-flow simulation, N-1 contingency analysis and 24-hour time-series operation.
- Evaluated EV charging, PV/storage integration, voltage stability, line loading and operational constraints.

District Heating Dispatch Optimisation

KTH Course Project | Python, PuLP/MILP | 2025

- Built a multi-asset heat dispatch optimisation model including CHP, heat pumps, storage, outages, electricity-price coupling and emissions costs.
- Compared cost and emissions objectives, seasonal operation, technology mix and KPI outcomes.

EDUCATION

M.Sc. in Sustainable Energy Engineering

KTH Royal Institute of Technology, Stockholm | 2023 - 2026

Relevant areas: Applied Heat and Power Technology, Numerical Heat Transfer in Energy Technology, Sustainable Power Generation, Modelling of Energy Systems, Energy Management, Practical Optimization of Energy Networks, Energy Systems for Sustainable Development, Energy in the Built Environment. Registered as a student for Autumn 2026.

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India | 2017 - 2021

Relevant areas: Thermodynamics, Heat and Mass Transfer, Fluid Mechanics, Thermal Engineering, Compressible Flow, Machine Design, CAD/CAE, Mechanical Design and Manufacturing.

TECHNICAL SKILLS

Thermo-fluids:	Thermodynamics; heat transfer; fluid mechanics; pressure drop; thermal limits; steady-state CFD; conjugate heat transfer; flow-performance interpretation.
Testing and instrumentation:	NI-DAQ; NI MAX; LabVIEW; thermocouples; static/dynamic pressure sensors; measurement-chain commissioning; signal checks; validation planning; test documentation.
Mechanical / CAD:	SolidWorks; Siemens NX; component design; assembly thinking; technical drawings; mechanical documentation; manufacturability awareness; design iteration.
Simulation and analysis:	ANSYS Fluent; ANSYS Meshing; SpaceClaim; Python post-processing; mesh checks; sensitivity studies; technical plots and KPI extraction.
Programming and data:	Python; pandas; NumPy; SciPy; matplotlib; PuLP; Git; Docker; Postman; Excel; structured debugging.
Energy systems:	Industrial energy efficiency; heat recovery; energy KPI/EnPI frameworks; district heating; renewable integration; power-flow project experience; sustainable power generation.
Documentation and quality mindset:	Technical reports; assumptions and limitations; validation summaries; root-cause analysis; clear English communication; cross-functional collaboration.

CERTIFICATIONS AND LANGUAGES

Certifications: SSG Entre / Industrial Safety, Sweden | Applied Computational Fluid Dynamics, Siemens | Schneider Electric - Assessing a Commercial Product's Sustainability | IBM Python for Data Science | Introduction to Git and GitHub | Digital Manufacturing and Design Technology, University at Buffalo.

Languages: English - professional proficiency | Malayalam - native | Hindi - working proficiency | Swedish - beginner, actively studying.

AVAILABILITY / ROLE INTEREST

Interested in Westinghouse Sweden opportunities in mechanical engineering, thermal hydraulics testing, development engineering, mechanical design/calculation support, quality/documentation-adjacent engineering, internships, summer work and early-career permanent roles. Available for Västerås-based opportunities, subject to role requirements and work authorization process.